

Trader Behavior Insights: Performance vs Bitcoin Market Sentiment

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1. Data & Methodology

Two datasets were merged on calendar date: **Hyperliquid historical trades** (211,224 rows, 32 accounts, 246 coins, May 2023 – May 2025) and the daily **Bitcoin Fear & Greed Index** (classification: Extreme Fear / Fear / Neutral / Greed / Extreme Greed). 211,218 of 211,224 trades (99.997%) matched a sentiment day. No nulls, duplicate rows, or duplicate index dates were found.

Performance metrics (PnL, win rate, payoff ratio, efficiency) use closing trades (Closed PnL $\neq$ 0; $n = 104,402$). **Behavioral metrics** (volume, positioning, sizing) use all trades. Total cohort closed PnL: **\$10.30M**. Headline differences were validated with Mann-Whitney U, two-proportion z, and Spearman rank tests. Analysis code: notebook_1.ipynb.

2. Key Findings

2.1 Profits live at the sentiment extremes, not in the middle

Closed PnL per active trading day: **Extreme Fear \$53.8k > Fear \$45.4k > Extreme Greed \$25.1k** $\approx$ Neutral \$23.5k > **Greed \$12.9k**. Plain Greed days are the weakest regime by every metric (median daily PnL only $\sim$ \$917). Because the relationship is U-shaped, daily PnL has essentially zero linear correlation with the raw index value (Spearman rho = -0.02 , $p = 0.63$) — a simple “more greed = more profit” story is wrong.

Regime	Trades	Total PnL	PnL / active day	Win rate	Payoff ratio	PnL per \$1k volume
Extreme Fear	10,406	\$739k	\$53.8k	76%	0.67	\$13.0
Fear	29,808	\$3,357k	\$45.4k	87%	0.97	\$14.0
Neutral	18,159	\$1,293k	\$23.5k	82%	0.92	\$12.8
Greed	25,176	\$2,150k	\$12.9k	77%	0.91	\$15.7
Extreme Greed	20,853	\$2,715k	\$25.1k	89%	1.34	\$46.8

Table 1 — Performance of closing trades by sentiment regime.

2.2 Extreme Greed is the highest-quality regime; Extreme Fear the highest-risk

Extreme Greed delivers the best win rate (89%), the only payoff ratio above 1 (1.34), and roughly **3× the efficiency of any other regime** (\$46.8 PnL per \$1k traded vs \$13–16 elsewhere). Extreme Fear is the mirror image: the lowest win rate (76%), average losses 1.5× average wins (\$257 vs \$173), a 5th-percentile trade of $-\$199$ (vs $-\$6$ in Extreme Greed), and only 64% of days net-profitable (vs 93%). Its large totals are brute-forced by volume — \$7.6M/day traded, $\sim 7.5\times$ the Neutral/Greed pace.

2.3 The long/short edge flips with the regime

In **Fear**, short closes earn the most: **\$208/trade at an 86% win rate** (longs: \$69). In **Extreme Greed**, longs dominate at **\$176/trade and a 91% win rate** while shorts collapse to \$29/trade. The single worst configuration in the dataset is shorting plain-Greed days (69% win rate, \$55/trade) — fading greed too early loses.

2.4 The cohort trades contrarian — and it paid

Long share of position-opening volume falls monotonically with sentiment: **75% long in Extreme Fear $\rightarrow$ 70% $\rightarrow$ 64% $\rightarrow$ 67% $\rightarrow$ 53% in Extreme Greed**. Traders also size up in Fear (avg trade \$7.8k) and size down in Extreme Greed (\$3.1k). The cumulative PnL curve confirms the payoff: roughly flat through the Greed-heavy

2024, then nearly the entire \$10.3M earned during the Fear/Extreme-Fear-heavy December 2024 – May 2025 stretch.

2.5 Hidden pattern: sentiment-regime transitions are the money days

Days when the sentiment classification flips average **\$38.4k vs \$19.7k** on stable days; days flipping *into* Fear average **\$60.0k** (n = 14 — directional evidence). Volatility at regime boundaries is opportunity for this cohort.

2.6 Statistical validation

Win rate is significantly higher on Fear days than Greed days (84.4% vs 82.5%; two-proportion z = 7.8, $p \approx 1e-14$), but mean per-trade PnL is **not** significantly different (Mann-Whitney $p = 0.53$). The regimes differ in *shape* — loss tails and payoff ratios — not in average trade outcome.

3. Visual Summary

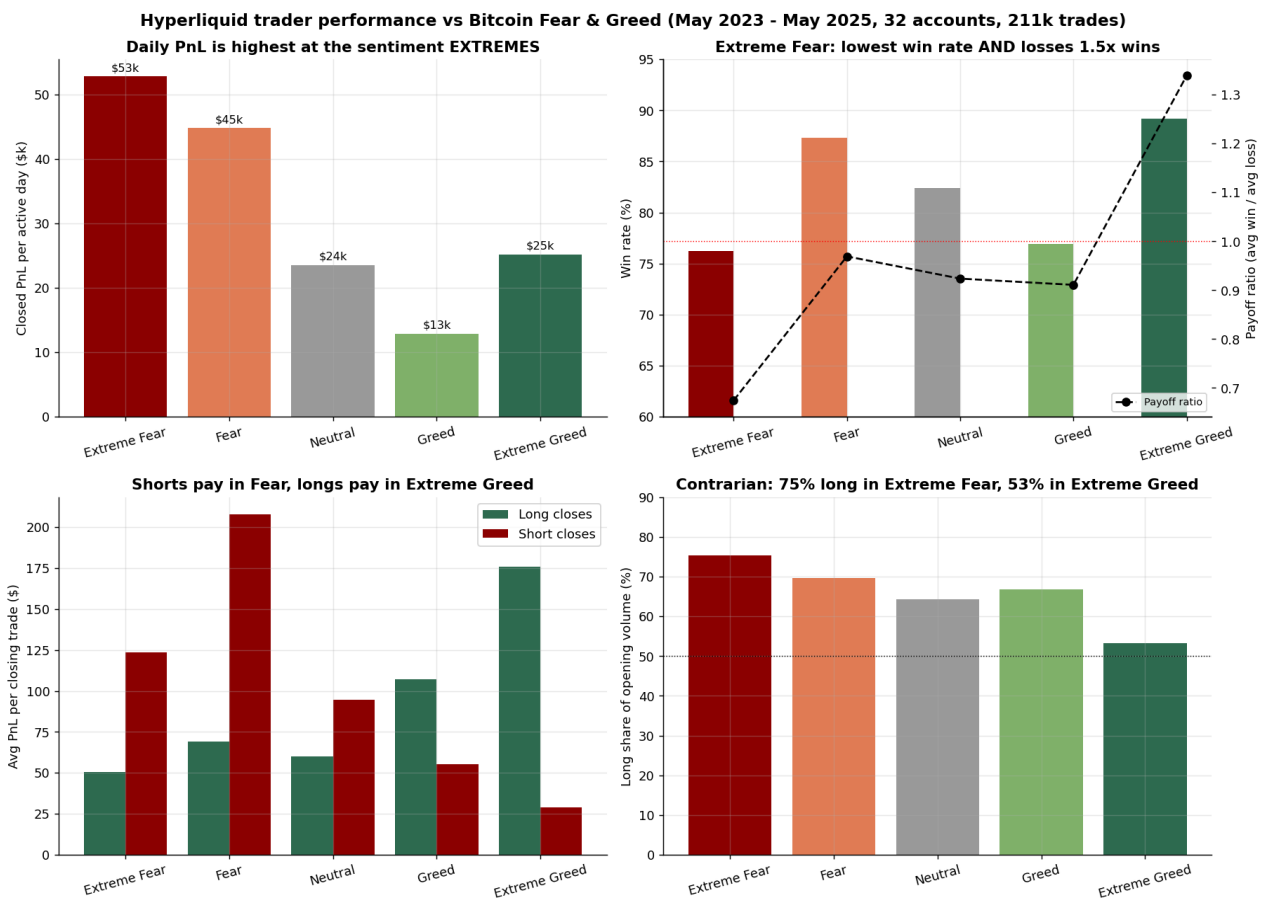


Figure 1 — Regime comparison: daily PnL, win rate & payoff ratio, long/short edge, positioning.

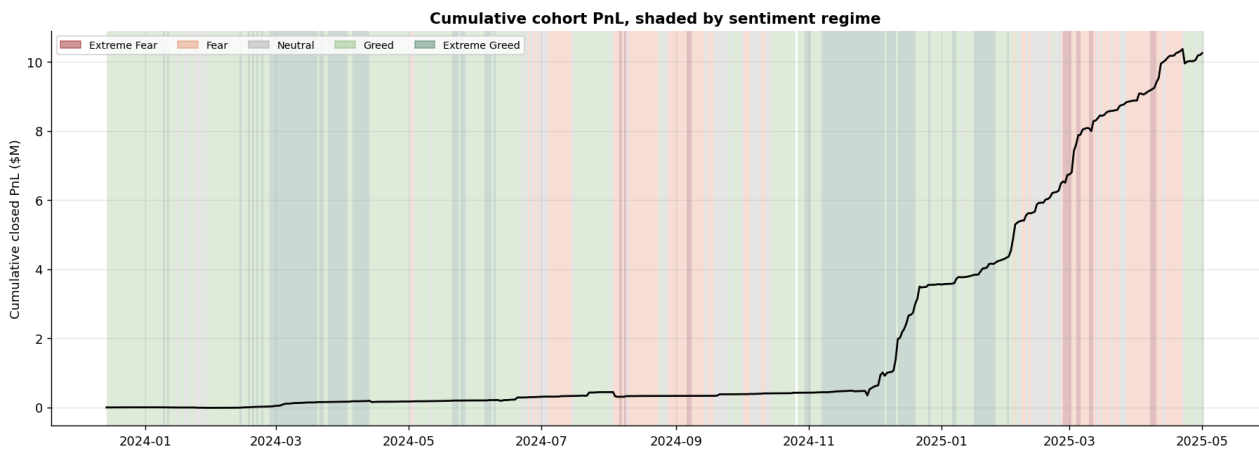


Figure 2 — Cumulative cohort PnL shaded by daily sentiment regime.

4. Strategy Implications

- **Trade the extremes, fade the middle.** Both sentiment extremes outperform; plain Greed days are the worst regime by every metric. Concentrate capital and attention on Extreme Fear, Fear, and Extreme Greed days.
- **Match direction to regime.** Short into Fear (\$208/trade, 86% win); ride longs in Extreme Greed (\$176/trade, 91% win); never short an Extreme Greed melt-up, and avoid fading plain Greed early.
- **Risk-manage Extreme Fear specifically.** It is the only regime where average losses exceed average wins (payoff 0.67) with a ~5× fatter left tail. Smaller size and tighter stops there convert the highest-\$ regime into a sustainable one.
- **Buy fear, trim into euphoria.** The cohort's monotonic contrarian positioning (75% → 53% long share) coincided with essentially all of its \$10.3M PnL.
- **Watch regime flips.** Sentiment classification changes — especially into Fear — marked the most profitable days (~2–3× stable days).

5. Limitations

32 accounts only; the top 3 hold 45.7% of total PnL (the regime pattern survives their exclusion — Extreme Greed win rate 90%, Extreme Fear 72% ex-top-3). Coin mix is confounded with time and regime (@107 earned almost everything in Greed regimes; HYPE/ETH/SOL/BTC mostly in Fear). Sentiment is BTC-based but applied to all coins. Closed PnL excludes fees (~3% of PnL in most regimes, 1% in Extreme Greed). This export has no leverage column, so leverage effects could not be isolated.